

Alessandro Sebastianelli, Ph.D.



About me

Alessandro Sebastianelli received the degree (cum laude) in Electronic Engineering from the University of Sannio in 2019, where he also obtained the Ph.D. degree in Information Technologies for Engineering in 2022. His expertise covers EO, AI, and quantum computing. He has coauthored a book and around 50 publications in reputed journals and international conferences, with an H-index of 13 and more than 775 citations. He has also managed and chaired sessions at leading workshops and international conferences. He received an IEEE award for one of the best theses in GRSS. He has been as a visiting researcher and later as a research fellow at the ESA Φ-lab for a total of six years, until 09/2025, where he was responsible for both the research line and industrial activities on quantum computing for EO. After a selective process, he was awarded ESA co-funding through the OSIP program to carry out research on AI-driven domain translation and data fusion for EO applications. He also worked for a few months as an EO Engineer at Titan4, focusing on structural monitoring based on SAR and optical satellite data. He currently leads the QC4EO working group within the QUEST IEEE GRSS Technical Committee and covers the role of Topical Associate Editor in IEEE TGRS. In 09/2025, he joined CMCC as a Scientist in the REMHI Division and PRESTO Unit.

Personal

Alessandro Sebastianelli
Nationality: Italian
YOB: 1995
Age: 29

[click me](#)

Areas of specialization

Remote Sensing Applications, Artificial Intelligence and Quantum Machine Learning

Interests

Earth Observation, Machine Learning, Quantum Technologies, Digital Technologies

Digital Skills

OS (Excellent), Programming (Excellent), Office suite (Excellent), CAD (limited), Internet (Excellent), Web-dev (limited), Multimedia (Excellent)

SHORT RESUMÉ

09/2025-today	Junior Scientist EURO-MEDITERRANEAN CENTER ON CLIMATE CHANGE CMCC · Caserta (IT) Supporting REMHI division and PRESTO Unit, as expert of Artificial Intelligence and Earth Observation.
10/2022-08/2025	Research Fellow in Quantum Computing for Earth Observation EUROPEAN SPACE AGENCY, Φ-LAB · Frascati (IT) Technical officer of QC4EO OSIP activities. Technical officer of ITT on QC4EO and Blockchain4EO. TEB expert evaluator. Research activities about quantum computing, quantum machine learning and quantum technologies applied to Earth Observation and space sector, with a focus on operability.
08/2020-10/2022	ESA OSIP EUROPEAN SPACE AGENCY, Φ-LAB · Frascati (IT) Co-funded activity on AI powered cross-modal adaptation techniques applied to S1 & S2 data.
09/2019-11/2019	Earth Observation Engineer TITAN4 · Rome (IT) structural monitoring based on SAR and optical satellite data. Use and validation of a tool for the measurement of displacements. The start-up was incubated in ESA BIC Lazio.
07/2019-11/2019	Visiting Researcher EUROPEAN SPACE AGENCY, Φ-LAB · Frascati (IT) Creation of EO datasets through the implementation of automatic tools for data retrieval and processing of those data with AI.

EDUCATION

12/2019-10/2022	Ph.D. in Information Technologies for Engineering 3 YEARS, GRADUATED · Un. of Sannio Satellite Remote Sensing through ML and Quantum Computing Techniques.
09/2017-10/2019	Master in Electronics Engineering 2 YEARS, 110/110 CUM LAUDE · Un. of Sannio SAR and Optical data fusion using AI. Generation of Sentinel-2 images from despeckled Sentinel-1 data.
09/2014-10/2017	Bachelor in Electronics Engineering 3 YEARS, 102/110 · Un. of Sannio Use of the Differential Interferometry on Sentinel images for the land displacements measurement. Ischia earthquake and comparison with INGV data.

CERTIFICATES & GRANTS ADDITIONAL INFO

2022 Global top 100 AI solutions for SDGs	Development experience: The complete record can be found on personal GitHub page, ESA Φ-lab page, UnisannioEOLab page (dev languages/environments - main: Python, bash and C, secondary: Java, javascript, HTML, MATLAB, Verilog). Scientific publications: 60 ~ 1000 citations, H-index of 15. A full list of my publications can be found here (click me). Talks: More than 10. A full list of talks I gave can be found here (click me). Supervising: More than 20 projects. A full list of supervising activities can be found here (click me). Events: A full list of events I co-organized can be found here (click me). IEEE: I am currently leading the QC4EO WG in IEEE GRSS QUEST and serving as TAS at TGRS
2022 Best UNICEF Research.	
2022 Wellcome Trust Support.	
2021 GRS 29-Italy 2020 award for the three best master's degree theses in geosciences and remote sensing	
2020 ESA OSIP co-founded Ph.D.	
2020 ESA Teamwork Excellence Award for the RACE Team through demonstrative excellent use of space for the benefit of society or the environment (by the ESA DG Josef Ashbacher).	

LANGUAGES

Italian	C2	mother tongue
English	C1	
French	A2	

PERSONAL SKILLS

High Experience and attitude in International Team Working. High attitude towards Problem Solving and Continuous Improvement. Results oriented, with ability of adapting to rapid changes. Hobbies: Photography, Playing Guitar.

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Date and signature:

Detailed info (up to April 14, 2026)

This part of my CV contains the more detailed information with respect to the first page. Most of these details can also be found, including images and/or documents, on my website <https://alessandrosebastianelli.github.io/>.

Awards & Certificates

Entity	Date	Description	URL
IEEE GRSS	2021	This award refers to my Master Thesis, but I was awarded during my Ph.D. and the research activities started during my master thesis have been carried on during my PhD and be significative to win the OSIP call.	click me
ESA OSIP	2021	Research Co-Sponsorship: AI Powered cross-modal adaptation techniques applied to Sentinel-1 and -2 data	click me
Unisannio	2022	Mention by the Rector of University of Sannio Prof. Gerardo Canfora about my research activities	click me
UNESCO/IRCAI	2022	Global top 100 AI solutions for SDGs Award 2022. This award refers to the Research Activity: Dengue Outbreaks Forecasting	click me
UNICEF	2022	Best UNICEF Research Award UNICEF. This award refers to the Research Activity: Dengue Outbreaks Forecasting	click me
WELLCOME TRUST	2022	Wellcome Trust Support. This award refers to the Research Activity: Dengue Outbreaks Forecasting	click me
ESA	2020	ESA Teamwork Excellence Award for the RACE Team through demostrating excellent use of space for the benefit of society or the environment (by the ESA DG Josef Ashbacher).	click me
IEEE IADF	2024	IEEE IADF School certificate of appreciation from IEEE GRSS for my contribution to the IEEE GRSS and IADF School on Computer Vision for Earth Observation	click me
Unisannio	2024	I have been awarded with a certificate of appreciation from Rector Gerardo Canfora (University of Sannio) for my contribution to the IEEE GRSS and IADF School on Computer Vision for Earth Observation.	click me
IRF	2024	Contribution as an external reviewer for the Icelandic Research Fund.	click me

Publications To limit the number of pages the list of publications is not displayed here, but can be accessed at the following clickable links: Scopus, WoS, Google Scholar. For a more structured view Homepage.

Talks and presentations

Organizer	Date	Description	URL
ESA	2020	φ-Week Side event organizer	click me
IEEE	2020	Paper presentation at IEEE IGARSS2020	click me
IEEE	2021	Paper presentation at MED2022	click me
IEEE	2021	Poster Presenter at GTTI2021	click me
MDPI	2022	Talk at IJGI Webinar The Rise of Artificial Intelligence for Space Applications	click me
Sapienza	2022	Talk at La Sapienza DIAG	click me
GTTI	2022	Poster Presenter at GTTI2022	click me
IEEE	2022	Papers presentation at IEEE MetroXRaine2022	click me
ECMWF	2022	Speaker at the ECMWF-ESA workshop	click me
Unisannio	2022	Speaker at CUSAS Workshop	click me
IEEE	2022	Co-chairing a session at IGARSS	click me
CINECA	2022	Talk at High Performance Computing and Quantum Computing 5th	click me
CINECA	2023	Talk and hands-on session at Introduction to Quantum Computing School	click me
Unipavia	2024	Seminar on Quantum Machine Learning for Earth Observation	click me
IEEE	2024	Talk on QC4EO at IEEE GRSS IADF School	click me
IEEE	2024	Speaker at IEEE GRSS IADF School	click me
Unibari	2024	Talk on QC4EO at Quantum 2024 - Summer School on Quantum Science and Technologies	click me
ESA	2025	Chairing a session on QC4EO at LP52	click me
ESA	2025	Agora session on GRSS/ESA collaboration at LPS	click me
IEEE	2025	Invited speaker at IEEE GRSS IADF School	click me

Supervising activities

Teaching support:

- Lessons of Remote Sensing for "Reti di Telecomunicazioni" course: how to download and elaborate satellite data with classic and AI-based techniques
- Support to Ph.D. course: Optical and Radar Remote Sensing
- Exam commissions: "Reti di Telecomunicazioni" and "Teoria ed Elaborazione dei Segnali"

Supervising activities:

- Jan 2021: Massachusetts Institute of Technology student co-supervisor University of Sannio, Benevento, Italy. I worked as a Tutor to support three MIT students in the development of their project: 1. Quantum Artificial Intelligence applied to Remote Sensing data and 2. Infrastructural monitoring using satellite data.
- Jan 2020: Massachusetts Institute of Technology student co-supervisor University of Sannio, Benevento, Italy ESA, ESRIN, Φ-Lab, Frascati, Rome, Italy. I worked as a Tutor to support two MIT students in the development of their project: 1. Land-slides detection based on artificial intelligence algorithms, 2. Infrastructural monitoring using satellite data and 3. Country development measurement using satellite data
- Jan 2019: Massachusetts Institute of Technology student co-supervisor University of Sannio, Benevento, Italy. I worked as a Tutor to support two MIT students in the development of their project: Landslides detection based on artificial intelligence algorithms.

Thesis co-relator:

- Master Thesis Co-Relator - Francesco Mauro. Master Degree in Electronic Engineering for Automation and Telecommunications, University of Sannio, Engineering Department, 2022. Climate Change Impact Evaluation on levels of water resources through deep learning techniques.
- Bachelor Thesis Co-Relator - Pietro Di Stasio. Bachelor Degree in Electronic Engineering for Automation and Telecommunications, University of Sannio, Engineering Department, 2022. Use of Sentinel-5P data for the early detection of volcanic eruptions through on-board Artificial Intelligence.

- Bachelor Thesis Co-Relator - Giovanni Pagnozzi. Bachelor Degree in Electronic Engineering for Automation and Telecommunications, University of Sannio, Engineering Department, 2022. Analysis of large strategic structures using the PyGMTSAR tool on Sentinel-1 data.
- Bachelor Thesis Co-Relator - Simona Reale. Bachelor Degree in Electronic Engineering for Automation and Telecommunications, University of Sannio, Engineering Department, 2021. Performance analysis of a new splitting method for datasets in machine learning models. Case study: detection of volcanic eruptions.
- Bachelor Thesis Co-Relator - Luigi Russo. Bachelor Degree in Electronic Engineering for Automation and Telecommunications, University of Sannio, Engineering Department, 2021. Development of a Machine Learning model based on the "categorical boosting" technique for the correlation between tropospheric NO2 and NO2 on the ground.
- Bachelor Thesis Co-Relator - Tony De Corso. Bachelor Degree in Electronic Engineering for Automation and Telecommunications, University of Sannio, Engineering Department, 2020. Application of DInSAR technique to high coherence satellite images for strategic infrastructure monitoring: Morandi Bridge.
- Bachelor Thesis Co-Relator - Morena Gismondi. Bachelor Degree in Electronic Engineering for Automation and Telecommunications, University of Sannio, Engineering Department, 2020. Use of Sentinel-5P data for the analysis of the correlation between NO2 levels and mobility data in areas with a high number of infections due to Covid-19. Case study: Lombardy Region.
- Bachelor Thesis Advisor - Luca Mignone. Bachelor Degree in Electronic Engineering for Automation and Telecommunications, University of Sannio, Engineering Department, 2020. Use of differential interferometry on Sentinel-1 images for the measurement of earthquake-induced ground displacements.
- Bachelor Thesis Advisor - Gianluca Di Cosmo. Bachelor Degree in Electronic Engineering for Automation and Telecommunications, University of Sannio, Engineering Department, 2020. Use of Sentinel-5P data for the analysis of the correlation between NO2 levels and the number of infections due to Covid-19. Case study: Wuhan area.
- Bachelor Thesis Co-Relator - Francesco Mauro. Bachelor Degree in Electronic Engineering for Automation and Telecommunications, University of Sannio, Engineering Department, 2020. Analysis of the correlation between Sentinel-5P data and epidemiological data. Case study: spread of Covid-19 in the Lombardy region.

Personal Skills

Languages

LANGUAGE	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken Interaction	Spoken Production	
Italian			mother tongue		
English	C1 Proficient	C1 Proficient	C1 Proficient	C1 Proficient	C1 Proficient
Fench	A2 Basic	A2 Basic	A2 Basic	A2 Basic	A2 Basic

Team working and problem solving: I worked in different research groups providing fundamental results and contributing to the resolution of technical problems with elasticity.

Working in a international environment: I participated in numerous international conferences and collaborations, giving several talks and posters. I am currently working in an international environment and so I am comfortable working in the international environment.

Communication and scientific marketing: I give a good number of talks and posters. I co-organized and conducted multiple international symposia and wrote several articles.

Digital Skills:

- Basic digital competence: Operating systems (Excellent), Programming languages (Excellent), Word processing (Excellent), Electronic spreadsheet (Excellent), CAD skills (Limited), Internet skills (Excellent), Web-site creation (Limited), Multimedia (Excellent),
- Programming languages known: Python, Java, C, Arduino, (C++) chiBios, Matlab, Simulink, Latex, C+ (good knowledge), P5.JS and JavaScript (basic knowledge), Processing, Labview, Ladder, SCADA, Verilog, VHDL, Mathematica, Pascal (basic knowledge).
- Software applications: Matlab, Labview, Eclipse, Word, Excel, PowerPoint, Arduino, Photoshop, SNAP, Mathematica, RSLogic, RSView, Unity, Quartus Prime, QGIS, Jupyter Notebook, Google Colaboratory, Google Earth Engine.

Additional information: IEEE Membership (3 years), IEEE GRSS member (3 years), IEEE Young Professional (1 year), IEEE AESS (2 years).

Hobbies: Photography, playing the guitar self-taught

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